Lecture 7: Eigenvectors and eigenvalues

- MAT201 Linear Algebra
- The directions a matrix does not turn

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Recap: matrices move vectors

- A 2×2 matrix A sends each vector v in the plane to a new vector Av.
- In general A can stretch, flip and turn a vector at the same time.

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Most vectors change direction

- Take A = [[2, 1], [1, 2]] and v = (1, 0).
- Then Av = (2, 1), which points in a different direction from v.
- So multiplying by A turned this vector.

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Definition

- A nonzero vector v is an eigenvector of A if $Av = \lambda v$ for some scalar λ.
- The scalar λ is the eigenvalue that belongs to v.
- Eigenvectors are never the zero vector.

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What Av = λv means geometrically

- Av lies on the same line through the origin as v: A does not turn v.
- If $\lambda > 1$ the eigenvector is stretched; if $0 < \lambda < 1$ it is shrunk.
- If $\lambda < 0$ it is flipped to point the other way; if $\lambda = 0$ it is sent to the zero vector.

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Example: a symmetric matrix

- Let A = [[2, 1], [1, 2]].
- A(1, 1) = (3, 3) = 3·(1, 1), so (1, 1) is an eigenvector with eigenvalue 3.
- A(1, −1) = (1, −1) = 1·(1, −1), so (1, −1) is an eigenvector with eigenvalue 1.

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Example: a diagonal matrix

- For A = [[2, 0], [0, 3]], $Ae_1 = 2e_1$ and $Ae_2 = 3e_2$.
- So e_1 and e_2 are eigenvectors, with eigenvalues 2 and 3.
- The eigenvalues of a diagonal matrix are its diagonal entries.

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Any multiple works too

- If v is an eigenvector with eigenvalue λ, so is cv for every scalar $c \neq 0$.
- $A(cv) = c(Av) = c(\lambda v) = \lambda(cv)$.
- So we usually give one simple eigenvector for each eigenvalue.

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Finding eigenvalues

- $Av = \lambda v$ is the same as $(A - \lambda I)v = 0$.
- This has a nonzero solution v exactly when $A - \lambda I$ is not invertible.
- So the eigenvalues are the solutions of $\det(A - \lambda I) = 0$, the characteristic equation.

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The characteristic polynomial of a 2×2 matrix

- For $A = [[a, b], [c, d]]$: $\det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc$.
- Expanded: $\lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$.
- Its roots are the eigenvalues of A.

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Worked example: eigenvalues

- $A = [[4, 1], [2, 3]]$.
- $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10$.
- $\lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$, so the eigenvalues are $\lambda = 2$ and $\lambda = 5$.

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Worked example: eigenvectors

- For each eigenvalue, solve $(A - \lambda I)v = 0$.
- $\lambda = 5$: $A - 5I = [[-1, 1], [2, -2]]$, so $-x + y = 0$ and $v = (1, 1)$.
- $\lambda = 2$: $A - 2I = [[2, 1], [2, 1]]$, so $2x + y = 0$ and $v = (1, -2)$.

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Check with trace and determinant

- The eigenvalues add up to the trace: $2 + 5 = 7 = 4 + 3$.
- The eigenvalues multiply to the determinant: $2 \cdot 5 = 10 = 4 \cdot 3 - 1 \cdot 2$.
- Use this check on every exam answer.

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Common mistakes

- Subtract λ only on the diagonal of A.
- In $(a - \lambda)(d - \lambda) - bc$ the product bc is subtracted, not added.
- An eigenvector must be nonzero; $v = 0$ always satisfies $Av = \lambda v$ and does not count.

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No real eigenvectors: rotations

- $R = [[0, -1], [1, 0]]$ turns every vector by 90 degrees.
- $\det(R - \lambda I) = \lambda^2 + 1$, which has no real roots.
- So a rotation by 90 degrees has no real eigenvalues and no real eigenvectors.

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Why eigenvectors matter

- Next lecture: diagonalization writes A using its eigenvectors.
- In a Markov chain, the long-run state is an eigenvector with eigenvalue 1.
- In PCA, the eigenvectors of the covariance matrix are the directions of largest variance.

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Summary

- $v \neq 0$ is an eigenvector of A if $Av = \lambda v$; λ is its eigenvalue.
- Eigenvalues solve $\det(A - \lambda I) = 0$.
- For each λ, solve $(A - \lambda I)v = 0$ to get its eigenvectors.
- Check: sum of eigenvalues = trace, product = determinant.

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